Cancer in British vegetarians: updated analyses of 4998 incident cancers in a cohort of 32,491 meat eaters, 8612 fish eaters, 18,298 vegetarians, and 2246 vegans

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Background: Vegetarian diets might affect the risk of cancer. Objective: The objective was to describe cancer incidence in vegetarians and nonvegetarians in a large sample in the United Kingdom. Design: This was a pooled analysis of 2 prospective studies including 61,647 British men and women comprising 32,491 meat eaters, 8612 fish eaters, and 20,544 vegetarians (including 2246 vegans). Cancer incidence was followed through nationwide cancer registries. Cancer risk by vegetarian status was estimated by using multivariate Cox proportional hazards models. Results: After an average follow-up of 14.9 y, there were 4998 incident cancers: 3275 in meat eaters (10.1%), 520 in fish eaters (6.0%), and 1203 in vegetarians (5.9%). There was significant heterogeneity between dietary groups in risks of the following cancers: stomach cancer [RRs (95% CIs) compared with meat eaters: 0.62 (0.27, 1.43) in fish eaters and 0.37 (0.19, 0.69) in vegetarians; P-heterogeneity = 0.006], colorectal cancer [RRs (95% CIs): 0.66 (0.48, 0.92) in fish eaters and 1.03 (0.84, 1.26) in vegetarians; P-heterogeneity = 0.033], cancers of the lymphatic and hematopoietic tissue [RRs (95% CIs): 0.96 (0.70, 1.32) in fish eaters and 0.64 (0.49, 0.84) in vegetarians; P-heterogeneity = 0.005], multiple myeloma [RRs (95% CIs): 0.77 (0.34, 1.76) in fish eaters and 0.23 (0.09, 0.59) in vegetarians; P-heterogeneity = 0.010], and all sites combined [RRs (95% CIs): 0.88 (0.80, 0.97) in fish eaters and 0.88 (0.82, 0.95) in vegetarians; P-heterogeneity = 0.0007]. Conclusion: In this British population, the risk of some cancers is lower in fish eaters and vegetarians than in meat eaters.